# Week 3: Matrix operations and orthogonal projection

Linear Algebra I

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*These notes were drafted with the help of ChatGPT from my handwritten notes and the discussion used to prepare the recitation.*

## 1. Matrix units

Last time we studied matrix multiplication and the relation between row rank and column rank. This week we use these ideas to explain how multiplication acts on rows and columns and how elementary operations produce an inverse. The homework also connects linear systems with determinants through Cramer's rule. We then study orthogonal projection, where the inner product determines the closest point on a line.

We write $M_{m,n}(\mathbb{R})$ for the space of real $m \times n$ matrices and $M_n(\mathbb{R})$ for the square matrices of order $n$. Vectors are columns unless stated otherwise. The identity matrix is $I_n$, the transpose of $A$ is $A^T$, and the Euclidean inner product is

$$(u \mid v) = u^T v = \sum_{i=1}^{n} u_i v_i, \qquad \|u\|^2 = u^T u.$$

In the September 20 homework, let $E_{k\ell} \in M_n(\mathbb{R})$ have entry 1 in row $k$, column $\ell$, and entry 0 everywhere else. We use the Kronecker delta

$$\delta_{ij} = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

Thus $(E_{k\ell})_{ij} = \delta_{ik}\delta_{j\ell}$.

For $A = (a_{ij}) \in M_n(\mathbb{R})$, the entry formula for multiplication gives

$$(E_{k\ell}A)_{ij} = \sum_{q=1}^{n} (E_{k\ell})_{iq} a_{qj} = \delta_{ik} a_{\ell j}.$$

Hence row $k$ of $E_{k\ell}A$ is row $\ell$ of $A$, and all its other rows are zero. Similarly,

$$(AE_{st})_{ij} = \sum_{q=1}^{n} a_{iq} (E_{st})_{qj} = a_{is}\delta_{jt}.$$

Column $t$ of $AE_{st}$ is column $s$ of $A$, and all its other columns are zero.

Applying the same calculation to two matrix units yields

$$E_{k\ell}E_{st} = \delta_{\ell s} E_{kt}.$$

In particular, this product is zero unless $\ell = s$.

The final part asks for $\operatorname{tr}((E_{\ell k})^T E_{st})$. Since $(E_{\ell k})^T = E_{k\ell}$ and $\operatorname{tr}(E_{kt}) = \delta_{kt}$, we obtain

$$\operatorname{tr}((E_{\ell k})^T E_{st}) = \operatorname{tr}(\delta_{\ell s} E_{kt}) = \delta_{\ell s}\delta_{kt}.$$

Here the trace is the sum of the diagonal entries. Keeping track of the indices before and after transposing is essential.

## 2. Rows and columns of a product

The preceding calculations are special cases of a general observation. Let

$$A \in M_{m,n}(\mathbb{R}), \qquad T = (t_{ki}) \in M_{s,m}(\mathbb{R}), \qquad S = (s_{j\ell}) \in M_{n,t}(\mathbb{R}).$$

Write the rows of $A$ as $r_1, \ldots, r_m$ and its columns as $c_1, \ldots, c_n$.

The $k$-th row of $TA$ is

$$\mathrm{row}_k(TA) = \sum_{i=1}^{m} t_{ki} r_i, \qquad 1 \le k \le s.$$

It is a linear combination of the rows of $A$, with coefficients supplied by row $k$ of $T$.

Likewise, the $\ell$-th column of $AS$ is

$$\mathrm{col}_\ell(AS) = \sum_{j=1}^{n} s_{j\ell} c_j, \qquad 1 \le \ell \le t.$$

It is a linear combination of the columns of $A$, with coefficients supplied by column $\ell$ of $S$.

Thus left multiplication forms combinations of rows, while right multiplication forms combinations of columns. This explains the distinction between elementary row operations and elementary column operations.

## 3. Cramer's rule and addition of right-hand sides

If $A \in M_n(\mathbb{R})$ is invertible, Cramer's rule expresses the solution of $Ax = b$ as

$$x_i = \frac{\det A_i}{\det A}, \qquad 1 \le i \le n,$$

where $A_i$ is obtained from $A$ by replacing its $i$-th column with $b$. The other columns remain in their original positions.

### Exercise 1.18(4)

Let $M = (m_1 \mid m_2)$ be an invertible $2 \times 2$ matrix, and suppose

$$Mx = b, \qquad Mx' = b'.$$

By linearity,

$$M(x + x') = b + b'.$$

Since the solution is unique, the solution corresponding to $b + b'$ is $x + x'$.

For the first coordinate, Cramer's rule gives

$$x_1 = \frac{\det(b \mid m_2)}{\det M}, \qquad x_1' = \frac{\det(b' \mid m_2)}{\det M}.$$

Applying the same rule to $b + b'$, and then multiplying by $\det M$, gives

$$\det(b + b' \mid m_2) = \det(b \mid m_2) + \det(b' \mid m_2).$$

The second coordinate similarly gives

$$\det(m_1 \mid b + b') = \det(m_1 \mid b) + \det(m_1 \mid b').$$

These identities explain additivity in one column when the other is fixed. Any nonzero fixed column can be completed to an invertible $2 \times 2$ matrix; if the fixed column is zero, the identities are immediate.

**Exercise 1.21(1)**

Consider

$$\begin{cases} 5x_1 + 7x_2 = 3, \\ 2x_1 + 4x_2 = 1. \end{cases}$$

The coefficient determinant and the two replacement determinants are

$$\Delta = \begin{vmatrix} 5 & 7 \\ 2 & 4 \end{vmatrix} = 6, \qquad \Delta_1 = \begin{vmatrix} 3 & 7 \\ 1 & 4 \end{vmatrix} = 5, \qquad \Delta_2 = \begin{vmatrix} 5 & 3 \\ 2 & 1 \end{vmatrix} = -1.$$

Consequently,

$$x_1 = \frac{5}{6}, \qquad x_2 = -\frac{1}{6}.$$

## 4. Computing inverses

For a $2 \times 2$ matrix with $ad - bc \neq 0$,

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Indeed, multiplying the two unscaled matrices in either order gives $(ad - bc)I_2$.

For larger matrices, elementary operations provide a systematic method. The distinction between multiplication on the left and on the right explains both algorithms.

### Row operations

Let $A \in M_n(\mathbb{R})$ be invertible. Suppose a sequence of elementary row operations reduces $A$ to $I_n$, and let $P$ be the product of the corresponding elementary matrices. Then

$$PA = I_n, \qquad P = A^{-1}.$$

Applying the same operations to $I_n$ therefore produces $A^{-1}$:

$$(A \mid I_n) \xrightarrow{\text{row operations}} (I_n \mid A^{-1}).$$

More generally, if $B \in M_{n,k}(\mathbb{R})$, the same operations give

$$(A \mid B) \xrightarrow{\text{row operations}} (I_n \mid A^{-1}B).$$

This simultaneously solves the matrix equation $AX = B$.

### Column operations

Suppose instead that column operations reduce $A$ to $I_n$, with combined right multiplier $Q$. Then

$$AQ = I_n, \qquad Q = A^{-1}.$$

Applying these operations to vertically stacked blocks gives

$$\begin{pmatrix} A \\ I_n \end{pmatrix} \xrightarrow{\text{column operations}} \begin{pmatrix} I_n \\ A^{-1} \end{pmatrix}.$$

If $B \in M_{k,n}(\mathbb{R})$, the same operations give

$$\begin{pmatrix} A \\ B \end{pmatrix} \xrightarrow{\text{column operations}} \begin{pmatrix} I_n \\ BA^{-1} \end{pmatrix}.$$

This solves $XA = B$. The horizontal arrangement is suited to row operations, and the vertical arrangement to column operations.

**Exercise 1(6) in §2.4**

The matrix in the homework is

$$A = \begin{pmatrix} 1 & 2 & 0 & -1 \\ -3 & -5 & 1 & 2 \\ 1 & 3 & 2 & -2 \\ 0 & 2 & 1 & -1 \end{pmatrix}.$$

To find its inverse, solve $Ax = b$ with an arbitrary right-hand side $b = (b_1, b_2, b_3, b_4)^T$. After the operations

$$R_2 \leftarrow R_2 + 3R_1, \qquad R_3 \leftarrow R_3 - R_1,$$
$$R_3 \leftarrow R_3 - R_2, \qquad R_4 \leftarrow R_4 - 2R_2, \qquad R_4 \leftarrow R_4 + R_3,$$

the system becomes

$$\begin{cases} x_1 + 2x_2 - x_4 = b_1, \\ x_2 + x_3 - x_4 = 3b_1 + b_2, \\ x_3 = -4b_1 - b_2 + b_3, \\ x_4 = -10b_1 - 3b_2 + b_3 + b_4. \end{cases}$$

Back substitution gives

$$\begin{aligned} x_1 &= -3b_1 - b_2 + b_3 - b_4, \\ x_2 &= -3b_1 - b_2 + b_4, \\ x_3 &= -4b_1 - b_2 + b_3, \\ x_4 &= -10b_1 - 3b_2 + b_3 + b_4. \end{aligned}$$

The coefficients of $b$ are the entries of the inverse:

$$A^{-1} = \begin{pmatrix} -3 & -1 & 1 & -1 \\ -3 & -1 & 0 & 1 \\ -4 & -1 & 1 & 0 \\ -10 & -3 & 1 & 1 \end{pmatrix}.$$

The same coefficients would appear in the right block when reducing $(A \mid I_4)$.

## 5. Orthogonal projection

The inner product adds geometric information to the matrix calculations. Fix $a \in \mathbb{R}^n$, $a \neq 0$. We want to decompose a vector $u$ as

$$u = p + r, \qquad p \in \mathbb{R}a, \qquad r \perp a.$$

Write $p = \lambda a$. The orthogonality condition is

$$0 = (u - \lambda a \mid a) = (u \mid a) - \lambda \|a\|^2.$$

It uniquely determines

$$\lambda = \frac{(u \mid a)}{\|a\|^2}.$$

Hence

$$p = \frac{(u \mid a)}{\|a\|^2} a, \qquad r = u - p.$$

Substitution verifies that $r \perp a$, so the decomposition exists and is unique. The vector $p$ is the orthogonal projection of $u$ onto the line $\mathbb{R}a$.

**The projection matrix**

Because $a$ and $u$ are columns,

$$(u \mid a) = a^T u, \qquad \|a\|^2 = a^T a.$$

The numerator $a^T u$ is a scalar. Moving that scalar to the right of $a$, and then using associativity, gives

$$p = \frac{a^T u}{a^T a}\, a = \frac{a(a^T u)}{a^T a} = \left(\frac{aa^T}{a^T a}\right) u.$$

Thus

$$p = Pu, \qquad P = \frac{aa^T}{a^T a}.$$

Here $aa^T$ is an $n \times n$ matrix, whereas $a^T a$ is the positive scalar $\|a\|^2$. The matrix $P$ depends only on the line of projection; $u$ is its input.

The matrix is symmetric:

$$P^T = \frac{(aa^T)^T}{a^T a} = P.$$

It also satisfies

$$P^2 = \frac{a(a^T a)a^T}{(a^T a)^2} = P.$$

The identity $P^2 = P$ says that projecting a second time does not change a vector already on the line.

**Example**

Take

$$u = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}, \qquad a = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}.$$

Then $a^T u = 9$ and $a^T a = 14$. Therefore,

$$p = \frac{1}{14}\begin{pmatrix} 9 \\ 18 \\ 27 \end{pmatrix}, \qquad r = \frac{1}{14}\begin{pmatrix} 19 \\ -32 \\ 15 \end{pmatrix}.$$

Indeed,

$$a^T r = \frac{19 - 64 + 45}{14} = 0.$$

The projection matrix is

$$P = \frac{1}{14}\begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{pmatrix}.$$

The orthogonal decomposition also identifies the closest point on the line. For any $\lambda \in \mathbb{R}$,

$$u - \lambda a = r + \left(\frac{9}{14} - \lambda\right) a.$$

The two terms are orthogonal, so

$$\|u - \lambda a\|^2 = \|r\|^2 + 14\left(\lambda - \frac{9}{14}\right)^2 = \frac{115}{14} + 14\left(\lambda - \frac{9}{14}\right)^2.$$

The minimum is attained only at $\lambda = 9/14$. Thus $p$ is the unique closest point, and the distance from $u$ to $\mathbb{R}a$ is $\sqrt{115/14}$.

## 6. Review: linear systems and matrix identities

### Columns, rows, and orthogonality

Let $A \in M_{m,n}(\mathbb{R})$, with columns $c_1, \dots, c_n$ and rows $r_1, \dots, r_m$. For $x \in \mathbb{R}^n$,

$$Ax = x_1 c_1 + \cdots + x_n c_n.$$

Thus $Ax = b$ asks whether $b$ can be expressed as a linear combination of the columns, with coefficients $x_1, \dots, x_n$.

From the row perspective,

$$Ax = b \quad \iff \quad r_i x = b_i \quad (1 \le i \le m).$$

Identifying the coordinates of row $r_i$ with the column $r_i^T$, this can also be written as $(r_i^T \mid x) = b_i$. In particular,

$$Ax = 0 \quad \iff \quad x \text{ is orthogonal to every row of } A.$$

This is the geometric interpretation of the homogeneous system.

### Rank and the number of solutions

The row rank and column rank agree:

$$\text{row rank of } A = \text{column rank of } A = \operatorname{rank} A.$$

The consistency criterion is

$$Ax = b \text{ is consistent} \quad \iff \quad \operatorname{rank} A = \operatorname{rank}(A \mid b).$$

If the system is consistent and $\operatorname{rank} A = r$, its number of free variables is $n - r$. It has a unique solution when $r = n$, and infinitely many solutions when $r < n$.

Can a real linear system have exactly two solutions? No. If $x \neq y$ are solutions, then for every $t \in \mathbb{R}$,

$$A(x + t(y - x)) = b + t(b - b) = b.$$

Since $y - x \neq 0$, different values of $t$ give different solutions.

More generally, if $x_0$ is one particular solution, then

$$\{x : Ax = b\} = x_0 + \ker A, \qquad \ker A = \{z : Az = 0\}.$$

Indeed, subtracting $x_0$ from any solution gives a homogeneous solution, and adding a homogeneous solution to $x_0$ gives a solution of the original system.

### Invertibility

For a square matrix $A \in M_n(\mathbb{R})$, the following conditions are equivalent:

(a) $A$ is invertible.

(b) $\operatorname{rank} A = n$.

(c) $Ax = 0$ has only the zero solution.

(d) The reduced row echelon form of $A$ is $I_n$.

(e) For every $b \in \mathbb{R}^n$, the system $Ax = b$ has a unique solution.

(f) $\det A \neq 0$.

When these conditions hold, the solution is $x = A^{-1}b$.

### Products, transposes, and inverses

For compatible matrix sizes, multiplication is associative:

$$(AB)C = A(BC).$$

It is generally not commutative: even when both products are defined, $AB$ and $BA$ need not be equal.

The transpose identities are

$$(A+B)^T = A^T + B^T, \qquad (AB)^T = B^T A^T.$$

For invertible square matrices,

$$(AB)^{-1} = B^{-1}A^{-1}, \qquad (A^{-1})^T = (A^T)^{-1}.$$

Taking either the transpose or the inverse of a product reverses the order of its factors.

### Diagonal, symmetric, and skew-symmetric matrices

If $D = \operatorname{diag}(d_1, \ldots, d_n)$, then

$$D^k = \operatorname{diag}(d_1^k, \ldots, d_n^k) \qquad (k \in \mathbb{Z}_{\geq 0}).$$

The matrix $D$ is invertible exactly when every $d_i \neq 0$, in which case

$$D^{-1} = \operatorname{diag}(d_1^{-1}, \ldots, d_n^{-1}).$$

A real square matrix is symmetric if $A^T = A$, and skew-symmetric if $A^T = -A$. Every real square matrix has the decomposition

$$A = \frac{A + A^T}{2} + \frac{A - A^T}{2}.$$

The first term is symmetric and the second is skew-symmetric. The decomposition is unique: if $A = S + K$ with $S^T = S$ and $K^T = -K$, then $A^T = S - K$; adding and subtracting these equations determines $S$ and $K$.

Finally, if $A$ and $B$ are symmetric matrices of the same order, then

$$(AB)^T = B^T A^T = BA.$$

Consequently, $AB$ is symmetric if and only if $AB = BA$.